

Causality Assessment of an Adverse Drug Reaction

Case Study: Ramesh Patel

Evaluate the relationship between a drug and an adverse reaction using standard causality assessment methods. Analyze case studies and classify reactions as Certain, Probable, or Possible.

1. Case Summary

Patient	Ramesh Patel
Age / Gender	32 years / Male
Diagnosis	Acute bacterial sinusitis
Suspected drug	Amoxicillin 500 mg orally, TID
Therapy started	01/03/2026
ADR onset	03/03/2026
Reaction	Generalized itching and skin rash over the chest
Action taken	Amoxicillin discontinued; antihistamines administered
Outcome	Complete recovery by 06/03/2026

2. Causality Assessment

The relationship is assessed as **Probable/Likely**. The reaction occurred approximately two days after amoxicillin was started, the symptoms are compatible with a hypersensitivity/allergic skin reaction, and the patient recovered after the suspected drug was withdrawn. No stronger alternative cause is provided. Rechallenge was not performed.

3. Assessment Against Standard Criteria

Criterion	Finding	Interpretation
Temporal relationship	ADR appeared 2 days after starting amoxicillin	Supports relationship
Clinical plausibility	Itching and rash are compatible with drug hypersensitivity	Supports relationship
Dechallenge	Symptoms resolved after withdrawal	Positive; supports relationship
Rechallenge	Not performed	Cannot strengthen causality
Alternative causes	None provided	No better alternative identified

4. Classification

Final classification: Probable/Likely. The evidence is stronger than a Possible relationship because there is a compatible time sequence and positive dechallenge, but it is not Certain because rechallenge was not performed and the case does not provide definitive evidence excluding all other causes.

5. Final Conclusion

The suspected drug is **amoxicillin**. The patient developed generalized itching and a chest rash two days after starting treatment for acute bacterial sinusitis. Amoxicillin was withdrawn and the patient recovered with antihistamine treatment. Therefore, the adverse reaction is classified as **Probable/Likely** based on the available case information.